

Lecture 2 Porbability

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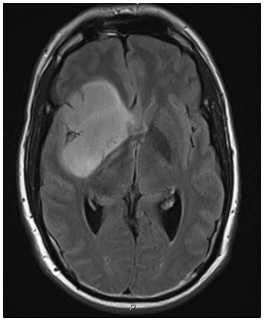
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Why Do We Need Probability in Machine Learning?



Brain MRI with a suspected tumor lesion

Suppose we train a model to detect whether a brain tumor is present from an MRI image.

Question

Can the model ever be perfectly certain about its prediction?

Uncertainty remains because

- training data are finite;
- the image may not contain all diagnostic information;
- measurements may be noisy or ambiguous.



Prediction Under Uncertainty

A deterministic view of machine learning is

$$\boxed{x \longrightarrow \widehat{y}}$$

where x denotes the observed features and $\widehat{y}$ is a prediction.

A probabilistic view is

$$\boxed{x \longrightarrow p(y \mid x)}$$

which describes our uncertainty about the possible value of y.

For example,

$$P(Y = \text{brain tumor} \mid X = x) = 0.82.$$

The model is not merely making a decision; it is quantifying uncertainty about that decision.

Two Sources of Uncertainty

The chapter distinguishes two important forms of uncertainty.

Epistemic uncertainty

Uncertainty due to limited knowledge.

- finite training data;
- imperfectly learned model;
- can generally be reduced by collecting more data.

Aleatoric uncertainty

Uncertainty arising from intrinsic variability or incomplete observations.

- noisy measurements;
- unobserved relevant variables;
- may remain even with unlimited training data.

Epistemic Uncertainty

Suppose we have only a small training sample

$$\mathcal{D} = \{(x_i, y_i) : i = 1, \ldots, n\}.$$

Different finite samples may lead us to learn somewhat different models.

Thus,

limited data $\implies$ uncertainty about the learned relationship

As n increases, we generally learn more about the underlying relationship:

more data $\implies$ less epistemic uncertainty.

"We do not know enough yet."

Aleatoric Uncertainty

Now imagine that the true outcome depends on variables

$$X_1, \quad X_2,$$

but we observe only X_1.
Even if the underlying mechanism is deterministic,

$$Y = f(X_1, X_2),$$

the relationship between Y and X_1 alone may appear noisy because X_2 is unobserved.
Thus,

$$\boxed{\text{unobserved information} \quad \Longrightarrow \quad \text{uncertainty in } Y \mid X_1}$$

"The observed input does not contain all the information."



The Brain-Tumor Example

Suppose we want to predict whether a brain tumor is present.

Using only an image,

$$p(\text{brain tumor} \mid \text{image})$$

may still have substantial uncertainty.

But suppose we also observe biopsy information:

$$p(\text{brain tumor} \mid \text{image}, \text{biopsy}).$$

The additional measurement may dramatically reduce uncertainty.

Richer information can reduce uncertainty.

Aleatoric Does Not Simply Mean “Unavoidable”

A useful distinction is:

More observations of the same limited features

may not eliminate aleatoric uncertainty.
But

collecting richer or different measurements

can sometimes reduce it.
For the brain tumor example:

$$\text{image only} \longrightarrow \text{image + biopsy}$$

provides more information about the underlying biological state.

Point Prediction vs. Distribution Prediction

Suppose our regression model predicts

$$\widehat{y} = 10.$$

What does this tell us about uncertainty?
A probabilistic model could instead say

$$Y \mid X = x \sim N(10, 2^2).$$

Now we know both

- the central prediction;
- how uncertain that prediction is.

Hence, a central object in probabilistic machine learning is

$$\boxed{p(y \mid x).}$$



Probability Appears Across Machine Learning

Classification

$$P(Y = k \mid X = x)$$

describes uncertainty about class membership.

Regression

$$p(y \mid x)$$

describes uncertainty about a continuous outcome.

Generative modeling

$$p(x) \qquad \text{or} \qquad p(x \mid z)$$

describes how data are generated.



Probability Appears Across Machine Learning

Parameter uncertainty

In Bayesian machine learning we may describe uncertainty about model parameters using

$$p(\boldsymbol{\theta} \mid \mathcal{D}).$$

Prediction

Probability allows us to combine uncertainty in parameters and observations to obtain predictive distributions such as

$$p(y_{\text{new}} \mid x_{\text{new}}, \mathcal{D}).$$

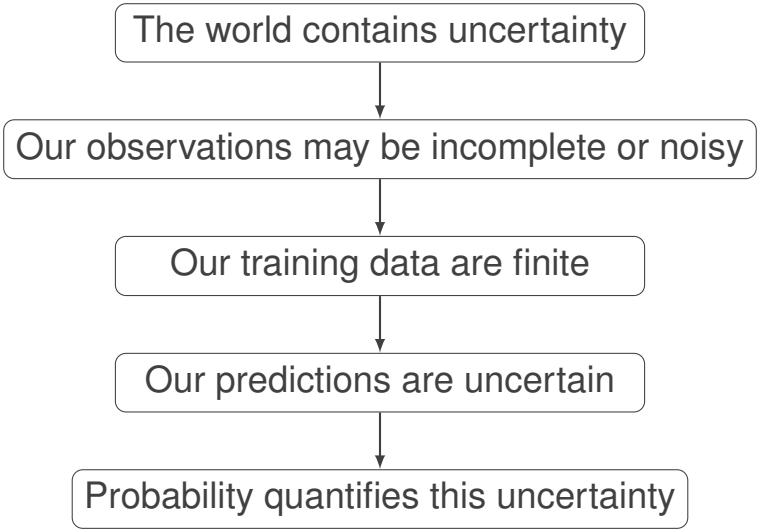
Why Probability?

Probability provides a consistent framework for

1. representing uncertainty;
2. combining different sources of information;
3. updating uncertainty when new information becomes available;
4. making predictions and decisions when information is incomplete.

Probability is the mathematical language of uncertainty.

The Big Picture



From Uncertainty to Probability Rules

We have motivated why probability is needed:

$$\boxed{x \longrightarrow p(y \mid x).}$$

But this raises the next question:

How do probabilities combine?

To answer this, we need three fundamental concepts:

joint probability	marginal probability	conditional probability.

Next: The Rules of Probability



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A Medical Screening Example

Consider a cancer screening test applied to a population.

Suppose that

$$p(\text{cancer}) = 0.01.$$

The test is not perfect:

- among people without cancer, 3% test positive;
- among people with cancer, 10% test negative.

Therefore, the test has

$$\boxed{\text{false-positive rate} = 0.03}$$

and

$$\boxed{\text{false-negative rate} = 0.10.}$$

A Medical Screening Example

The screening example raises two different questions.

Question 1

If we screen a randomly selected person, what is the probability that the test result is positive?

Question 2

If a person receives a positive result, what is the probability that the person actually has cancer?

These questions require us to distinguish among

joint probability	marginal probability	conditional probability.

The Sum and Product Rules

Let X and Y be two discrete random variables. Suppose X can take the values $x_1, \ldots, x_L$, and Y can take the values $y_1, \ldots, y_M$.
Consider N observations of the pair (X, Y). Define

$$n_{ij} = \text{number of observations with } X = x_i \text{ and } Y = y_j.$$

Also define

$$c_i = \sum_{j=1}^{M} n_{ij}, \qquad r_j = \sum_{i=1}^{L} n_{ij}.$$

Here, c_i is a column total and r_j is a row total.

The Sum and Product Rules

The N observations can be organized as follows.

	$X = x_1$	$X = x_2$	$\cdots$	$X = x_L$	Total
$Y = y_1$	n_{11}	n_{21}	$\cdots$	n_{L1}	r_1
$Y = y_2$	n_{12}	n_{22}	$\cdots$	n_{L2}	r_2
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
$Y = y_M$	n_{1M}	n_{2M}	$\cdots$	n_{LM}	r_M
Total	c_1	c_2	$\cdots$	c_L	N

The cell count n_{ij} corresponds to the joint event

$$X = x_i \qquad \text{and} \qquad Y = y_j.$$

The Sum Rule

The **joint probability** of $X = x_i$ and $Y = y_j$ is

$$\boxed{p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}.}$$

The **marginal probability** of $X = x_i$ is

$$\boxed{p(X = x_i) = \frac{c_i}{N}.}$$

Since every observation belongs to one of the L columns, $\sum_{i=1}^{L} c_i = N$. Therefore,

$$\sum_{i=1}^{L} p(X = x_i) = \sum_{i=1}^{L} \frac{c_i}{N} = 1.$$



The Sum Rule

The column total c_i is obtained by adding the counts within that column:

$$c_i = \sum_{j=1}^{M} n_{ij}.$$

Dividing both sides by N, $\frac{c_i}{N} = \sum_{j=1}^{M} \frac{n_{ij}}{N}$. Therefore,

$$\boxed{p(X = x_i) = \sum_{j=1}^{M} p(X = x_i, Y = y_j).}$$

More compactly,

$$\boxed{p(X) = \sum_{Y} p(X, Y).}$$

This is the **sum rule of probability**.



The Product Rule

Now consider only observations for which $X = x_i$. There are c_i such observations. Among them, n_{ij} also satisfy $Y = y_j$.
Therefore, the conditional probability is

$$\boxed{p(Y = y_j \mid X = x_i) = \frac{n_{ij}}{c_i}.}$$

The conditional probabilities are normalized:

$$\sum_{j=1}^{M} p(Y = y_j \mid X = x_i) = 1.$$

Conditioning changes the reference population from all N observations to the c_i observations satisfying $X = x_i$.

The Product Rule

We have

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}.$$

Insert the column total c_i:

$$\frac{n_{ij}}{N} = \frac{n_{ij}}{c_i}\frac{c_i}{N} = p(Y = y_j \mid X = x_i)p(X = x_i).$$

Therefore,

$$\boxed{p(X = x_i, Y = y_j) = p(Y = y_j \mid X = x_i)p(X = x_i).}$$

More compactly,

$$\boxed{p(X, Y) = p(Y \mid X)p(X).}$$

This is the **product rule of probability**.

The Two Fundamental Rules

The two fundamental rules of probability are

$$\boxed{p(X) = \sum_{Y} p(X, Y)}$$

and

$$\boxed{p(X, Y) = p(Y \mid X)p(X).}$$

Their interpretations are:

- $p(X, Y)$: the probability of X and Y;
- $p(Y \mid X)$: the probability of Y given X;
- $p(X)$: the marginal probability of X.

Sum over alternatives; multiply along conditions.

Bayes' Theorem

The product rule gives

$$p(X, Y) = p(Y \mid X)p(X).$$

Reversing the roles of X and Y,

$$p(Y, X) = p(X \mid Y)p(Y).$$

Then,

$$p(Y \mid X)p(X) = p(X \mid Y)p(Y).$$

Therefore,

$$\boxed{p(Y \mid X) = \frac{p(X \mid Y)p(Y)}{p(X)}.}$$

This is **Bayes' theorem**.

The Normalization Constant

Bayes' theorem is

$$p(Y \mid X) = \frac{p(X \mid Y)p(Y)}{p(X)}.$$

Using the sum and product rules,

$$p(X) = \sum_Y p(X, Y) = \sum_Y p(X \mid Y)p(Y).$$

Therefore,

$$\boxed{p(Y \mid X) = \frac{p(X \mid Y)p(Y)}{\sum_{Y'} p(X \mid Y')p(Y')}.}$$

The denominator $p(X)$ ensures that

$$\sum_Y p(Y \mid X) = 1.$$

Medical Screening Revisited

Let

$$C = \begin{cases} 1, & \text{cancer}, \\ 0, & \text{no cancer}, \end{cases} \qquad T = \begin{cases} 1, & \text{positive test}, \\ 0, & \text{negative test}. \end{cases}$$

The prior probabilities are

$$p(C = 1) = 0.01, \qquad p(C = 0) = 0.99.$$

The conditional test probabilities are

$$p(T = 1 \mid C = 1) = 0.90, \qquad p(T = 0 \mid C = 1) = 0.10,$$

and

$$p(T = 1 \mid C = 0) = 0.03, \qquad p(T = 0 \mid C = 0) = 0.97.$$

Medical Screening Revisited

A positive test can occur in two mutually exclusive ways:

$$C = 1,\ T = 1$$

or

$$C = 0,\ T = 1.$$

Using the sum and product rules,

$$\begin{aligned} p(T = 1) &= p(T = 1 \mid C = 1)p(C = 1) \\ &\quad + p(T = 1 \mid C = 0)p(C = 0) \\ &= (0.90)(0.01) + (0.03)(0.99) \\ &= 0.0387. \end{aligned}$$

Therefore

Medical Screening Revisited

We now want to reverse the conditioning direction:

$$p(C = 1 \mid T = 1).$$

Using Bayes' theorem,

$$\begin{aligned} p(C = 1 \mid T = 1) &= \frac{p(T = 1 \mid C = 1)p(C = 1)}{p(T = 1)} \\ &= \frac{(0.90)(0.01)}{0.0387} = \frac{90}{387} \approx 0.23. \end{aligned}$$

Thus,

$$\boxed{p(C = 1 \mid T = 1) \approx 23\%.}$$

Why Is the Answer Only 23%?

Imagine screening $10{,}000$ people.

	$T = 1$	$T = 0$	Total
$C = 1$	90	10	100
$C = 0$	297	9,603	9,900
Total	387	9,613	10,000

Among the 387 people who test positive, 90 have cancer. Therefore,

$$p(C = 1 \mid T = 1) = \frac{90}{387} \approx 0.23.$$

Prior and Posterior Probabilities

Before observing the test result, our knowledge about cancer is represented by

$$\underbrace{p(C)}_{\text{prior probability}} = 0.01.$$

After observing the test result, our updated knowledge is represented by

$$\underbrace{p(C \mid T)}_{\text{posterior probability}} \approx 0.23.$$

Bayes' theorem combines prior information with new evidence.



Independent Variables

Two random variables X and Y are independent if

$$\boxed{p(X,Y) = p(X)p(Y).}$$

An example is a sequence of independent coin flips: observing one flip does not change the probability distribution of the next flip.



An Uninformative Screening Test

Suppose the test result T were independent of cancer status C.
Then

$$p(T \mid C) = p(T).$$

Using Bayes' theorem,

$$p(C \mid T) = \frac{p(T \mid C)p(C)}{p(T)} = \frac{p(T)p(C)}{p(T)}.$$

Therefore,

$$\boxed{p(C \mid T) = p(C).}$$

The test result would not change our probability of cancer.
Such a test would be useless because it would provide no information about cancer status.



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From Probabilities to Densities

For a discrete random variable, we can assign probability to an exact value:

$$p(X = x).$$

For a continuous random variable, an exact value has probability zero:

$$P(X = x) = 0.$$

Instead, probability is assigned to an **interval**:

$$P(a < X < b).$$

Probability density

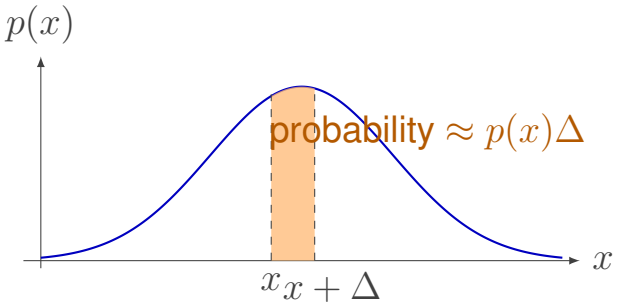
A density describes how probability is distributed locally along the real line.

From Probabilities to Densities

Let $p(x)$ be the probability density of a continuous variable X.

For a very small interval of width Δ,

$$P\big(x < X < x + \Delta\big) \approx p(x)\,\Delta.$$



From Probabilities to Densities

The probability that X lies between a and b is

$$\boxed{P(a < X < b) = \int_a^b p(x)\,dx.}$$

A valid probability density must satisfy

$$\boxed{p(x) \geq 0}$$

and

$$\boxed{\int_{-\infty}^{\infty} p(x)\,dx = 1.}$$

Density is not probability

The value $p(x)$ may exceed 1. Only an integrated probability must lie between 0 and 1.

The Cumulative Distribution Function

The cumulative distribution function, or CDF, is

$$\boxed{P(z) = P(X \leq z) = \int_{-\infty}^{z} p(x)\,dx.}$$

It accumulates probability from the left up to z.
If P is differentiable, then

$$\boxed{p(x) = \frac{dP(x)}{dx}.}$$

Therefore,

$$P(a < X \leq b) = P(b) - P(a).$$

Two views of the same distribution

$p(x)$ describes local probability density, whereas $P(x)$ describes accumulated probability.

Joint Densities for Several Variables

For a vector of continuous variables

$$\boldsymbol{x} = (x_1, \ldots, x_D)^\mathsf{T},$$

the joint density is

$$p(\boldsymbol{x}) = p(x_1, \ldots, x_D).$$

The probability of a region $\mathcal{A}$ is

$$\boxed{P(\boldsymbol{X} \in \mathcal{A}) = \int_{\mathcal{A}} p(\boldsymbol{x})\, d\boldsymbol{x}.}$$

The density must satisfy

$$p(\boldsymbol{x}) \geq 0, \qquad \int p(\boldsymbol{x})\, d\boldsymbol{x} = 1.$$

Discrete and Continuous Variables Together

Suppose C is discrete and X is continuous. A mixed joint density can be written as

$$\boxed{p(c, x) = P(C = c)\, p(x \mid C = c).}$$

Its interpretation is

$$P\big(C = c,\ x < X < x + \Delta\big) \approx p(c, x)\, \Delta.$$

The two marginal distributions use different operations:

$$P(C = c) = \int p(c, x)\, dx,$$

whereas

$$p(x) = \sum_{c} p(c, x).$$

Rule of thumb

Sum over discrete variables and integrate over continuous variables.

Probability Rules for Continuous Variables

Sum rule: $$\boxed{p(x) = \int p(x, y)\, dy},$$

Product rule: $$\boxed{p(x, y) = p(y \mid x)p(x)},$$

Bayes' theorem: $$\boxed{p(y \mid x) = \frac{p(x \mid y)p(y)}{p(x)}}.$$

The denominator in Bayes' theorem is

$$p(x) = \int p(x \mid y)p(y)\, dy.$$

Why Does a Sum Become an Integral?

Divide the real line into small intervals of width Δ. Treat each interval as one discrete outcome:

$$P(X \text{ lies in bin } i) \approx p(x_i)\Delta.$$

Then an interval probability is approximated by

$$P(a < X < b) \approx \sum_{x_i \in (a,b)} p(x_i)\Delta.$$

As the bins become arbitrarily narrow,

$$\boxed{\sum_{x_i \in (a,b)} p(x_i)\Delta \longrightarrow \int_a^b p(x)\,dx.}$$

Uniform Distribution

A constant density over the whole real line cannot be normalized and is therefore **improper**.
A constant density over a finite interval (c, d) is valid:

$$p(x) = \begin{cases} \dfrac{1}{d-c}, & c < x < d, \\ 0, & \text{otherwise.} \end{cases}$$

Its total area is

$$\int_c^d \frac{1}{d-c}\, dx = 1.$$

All intervals of the same length inside (c, d) have the same probability.



Exponential and Laplace Distributions

The exponential density is

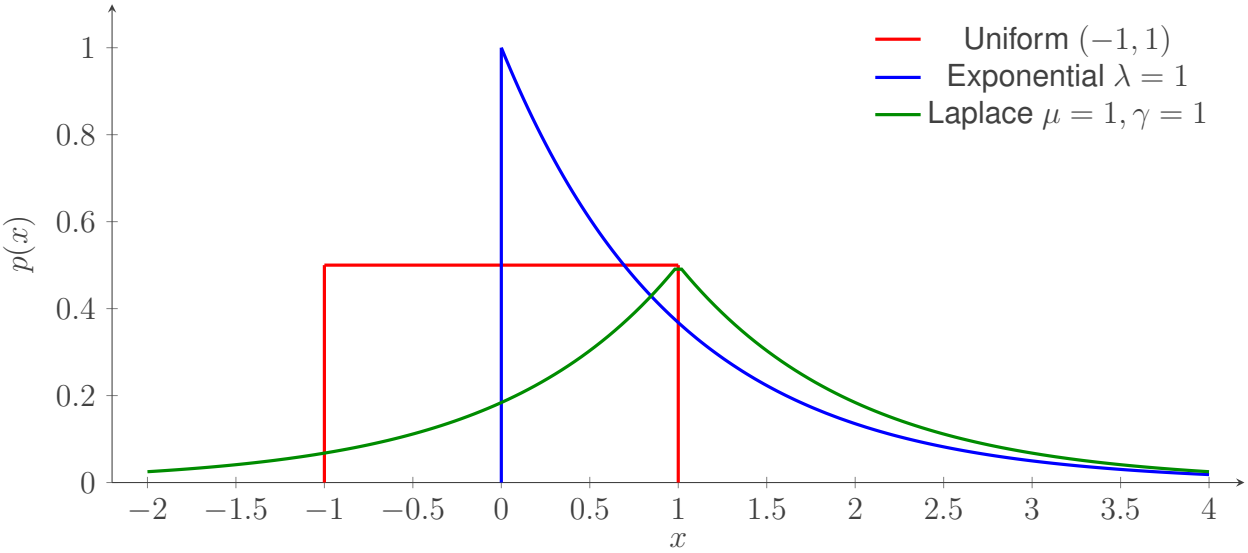
$$p(x \mid \lambda) = \lambda e^{-\lambda x}, \qquad x \geq 0.$$

It is often used for nonnegative quantities such as waiting times.
The Laplace density is

$$p(x \mid \mu, \gamma) = \frac{1}{2\gamma} \exp\left(-\frac{|x-\mu|}{\gamma}\right).$$

It is symmetric about μ and has a sharp peak and heavier tails than a Gaussian density.

Three Example Densities



Different density shapes encode different assumptions about where observations are likely to occur.

The Dirac Delta Function

The Dirac delta is written as

$$\boxed{p(x \mid \mu) = \delta(x - \mu).}$$

Informally, it is an infinitely narrow, infinitely tall spike at $x = \mu$ with unit area:

$$\delta(x - \mu) = 0 \qquad \text{for } x \neq \mu,$$

and

$$\int_{-\infty}^{\infty} \delta(x - \mu)\, dx = 1.$$

For a suitable function f,

$$\int f(x)\delta(x - \mu)\, dx = f(\mu).$$

Interpretation

All probability mass is concentrated at the single value μ.



The Empirical Distribution

Given observations

$$\mathcal{D} = \{x_1, \ldots, x_N\},$$

the empirical distribution is

$$\boxed{p(x \mid \mathcal{D}) = \frac{1}{N} \sum_{n=1}^{N} \delta(x - x_n).}$$

It places weight $1/N$ at each observed data point.
Its integral is

$$\int p(x \mid \mathcal{D})\, dx = \frac{1}{N} \sum_{n=1}^{N} 1 = 1.$$

This distribution represents the observed sample without imposing a smooth parametric form.



Expectation as a Probability-Weighted Average

The expectation of a function $f(X)$ is its probability-weighted average.
For a discrete variable,

$$\boxed{\mathbb{E}[f(X)] = \sum_x p(x) f(x).}$$

For a continuous variable,

$$\boxed{\mathbb{E}[f(X)] = \int p(x) f(x)\, dx.}$$

Important special cases include

$$\mathbb{E}[X], \qquad \mathbb{E}[X^2], \qquad \mathbb{E}[g(X)].$$



Estimating an Expectation from Data

Suppose $x_1, \ldots, x_N$ are drawn from $p(x)$.
Then

$$\boxed{\mathbb{E}[f(X)] \simeq \frac{1}{N} \sum_{n=1}^{N} f(x_n).}$$

The approximation becomes exact in the limit $N \to \infty$, under suitable conditions.

Partial and Conditional Expectations

For a function of two variables, the subscript specifies which variable is averaged over:

$$\mathbb{E}_X[f(X, Y)].$$

After averaging over X, the result is generally still a function of Y.
The conditional expectation of $f(X)$ given $Y = y$ is

$$\boxed{\mathbb{E}_X[f(X) \mid Y = y] = \int p(x \mid y) f(x)\, dx.}$$

For a discrete X, replace the integral with a sum:

$$\mathbb{E}_X[f(X) \mid Y = y] = \sum_x p(x \mid y) f(x).$$

Variance

Variance measures spread around the mean:

$$\boxed{\operatorname{var}[f(X)] = \mathbb{E}\left[(f(X) - \mathbb{E}[f(X)])^2\right].}$$

Expanding the square gives the computational identity

$$\boxed{\operatorname{var}[f(X)] = \mathbb{E}[f(X)^2] - \mathbb{E}[f(X)]^2.}$$

In particular,

$$\boxed{\operatorname{var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2.}$$

Variance is always nonnegative and is measured in squared units.



Covariance

Covariance measures whether two variables tend to vary together:

$$\boxed{\operatorname{cov}[X, Y] = \mathbb{E}\left[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\right].}$$

Equivalently,

$$\boxed{\operatorname{cov}[X, Y] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].}$$

- Positive covariance: large values tend to occur together.
- Negative covariance: one tends to be large when the other is small.
- Zero covariance: no linear co-variation.

Important distinction

Independence implies zero covariance, but zero covariance does not generally imply independence.



Covariance Matrices

For random vectors $\boldsymbol{X}$ and $\boldsymbol{Y}$,

$$\boxed{\operatorname{cov}[\boldsymbol{X}, \boldsymbol{Y}] = \mathbb{E}\left[(\boldsymbol{X} - \mathbb{E}[\boldsymbol{X}])(\boldsymbol{Y} - \mathbb{E}[\boldsymbol{Y}])^{\mathsf{T}}\right].}$$

Equivalently,

$$\operatorname{cov}[\boldsymbol{X}, \boldsymbol{Y}] = \mathbb{E}[\boldsymbol{X}\boldsymbol{Y}^{\mathsf{T}}] - \mathbb{E}[\boldsymbol{X}]\mathbb{E}[\boldsymbol{Y}]^{\mathsf{T}}.$$

For one vector,

$$\operatorname{cov}[\boldsymbol{X}] \equiv \operatorname{cov}[\boldsymbol{X}, \boldsymbol{X}].$$

Diagonal entries are component variances, whereas off-diagonal entries are pairwise covariances.



Quick Check

Let X have density

$$p(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find

$$P(X < 1/2)$$

and

$$\mathbb{E}[X].$$

Quick Check

Let X have density

$$p(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find

$$P(X < 1/2)$$

and

$$\mathbb{E}[X].$$

$$P(X < 1/2) = \int_0^{1/2} 2x \, dx = \frac{1}{4},$$

$$\mathbb{E}[X] = \int_0^1 x(2x) \, dx = \frac{2}{3}.$$



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Gaussian Distribution

For a real-valued variable x, the Gaussian density is

$$\boxed{\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}.}$$

It has two parameters:

$$\underbrace{\mu}_{\text{mean}} \quad \text{and} \quad \underbrace{\sigma^2}_{\text{variance}}.$$

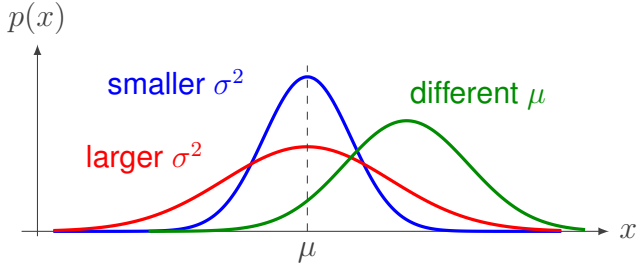
The standard deviation is

$$\sigma = \sqrt{\sigma^2},$$

and the precision is

$$\beta = \frac{1}{\sigma^2}.$$

Gaussian Distribution



- Changing μ shifts the density left or right.
- Increasing σ^2 makes the density wider and flatter.
- Decreasing σ^2 makes the density narrower and taller.

Gaussian Distribution

The Gaussian density is positive for every finite x:

$$\boxed{\mathcal{N}(x \mid \mu, \sigma^2) > 0.}$$

It is also normalized:

$$\boxed{\int_{-\infty}^{\infty} \mathcal{N}(x \mid \mu, \sigma^2)\, dx = 1.}$$

Therefore, it satisfies the two requirements for a probability density.

Although the Gaussian assigns positive density to every real value, its density decreases rapidly as x moves away from μ.



Mean and Second Moment

The expected value of a Gaussian variable is

$$\boxed{\mathbb{E}[X] = \int_{-\infty}^{\infty} x\,\mathcal{N}(x \mid \mu, \sigma^2)\,dx = \mu.}$$

The second moment is

$$\boxed{\mathbb{E}[X^2] = \int_{-\infty}^{\infty} x^2\mathcal{N}(x \mid \mu, \sigma^2)\,dx = \mu^2 + \sigma^2.}$$

These are called the first-order and second-order moments of the distribution.



Variance and Mode

Using

$$\mathrm{var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2,$$

we obtain

$$\begin{aligned}\mathrm{var}(X) &= (\mu^2 + \sigma^2) - \mu^2 \\ &= \boxed{\sigma^2}.\end{aligned}$$

The maximum of a density is called its **mode**. For a Gaussian,

$$\boxed{\text{mean} = \text{median} = \text{mode} = \mu.}$$

This equality follows from the symmetry of the Gaussian density around μ.



Estimating a Gaussian from Data

Suppose we observe

$$\mathcal{D} = \{x_1, \ldots, x_N\},$$

where the observations are independently drawn from

$$X_n \sim \mathcal{N}(\mu, \sigma^2).$$

The observations are **independent and identically distributed**, or i.i.d.

The parameters μ and σ^2 are unknown. We want to estimate them from the observed data.

Density estimation

Estimating an unknown probability distribution from a finite data set is called density estimation.



The Gaussian Likelihood Function

Because the observations are independent, their joint density is the product of their individual densities:

$$p(\mathcal{D} \mid \mu, \sigma^2) = \prod_{n=1}^{N} \mathcal{N}(x_n \mid \mu, \sigma^2).$$

When viewed as a function of the parameters μ and σ^2, this is the **likelihood function**.

Maximum likelihood

Choose the parameter values that make the observed data most likely:

$$(\widehat{\mu}, \widehat{\sigma}^2) = \arg\max_{\mu, \sigma^2} p(\mathcal{D} \mid \mu, \sigma^2).$$



Why Use the Log Likelihood?

The likelihood is a product:

$$p(\mathcal{D} \mid \mu, \sigma^2) = \prod_{n=1}^{N} \mathcal{N}(x_n \mid \mu, \sigma^2).$$

Taking logarithms converts the product into a sum:

$$\log p(\mathcal{D} \mid \mu, \sigma^2) = \sum_{n=1}^{N} \log \mathcal{N}(x_n \mid \mu, \sigma^2).$$

This is useful because:

- sums are easier to differentiate than products;
- the logarithm does not change the maximizer;
- it avoids numerical underflow from multiplying many small numbers.

The Gaussian Log Likelihood

Substituting the Gaussian density gives

$$\log p(\mathcal{D} \mid \mu, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{n=1}^{N} (x_n - \mu)^2 - \frac{N}{2} \log \sigma^2 - \frac{N}{2} \log(2\pi).$$

The first term measures how far the observations are from μ.
The remaining terms penalize inappropriate choices of the variance.

Maximizing the log likelihood balances fitting the observations closely against maintaining a properly normalized density.

Maximum Likelihood Estimates

Maximizing the log likelihood with respect to μ gives

$$\boxed{\mu_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N} x_n.}$$

Thus, the maximum likelihood estimate of the mean is the sample mean.
Maximizing with respect to σ^2 gives

$$\boxed{\sigma^2_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}(x_n - \mu_{\mathrm{ML}})^2.}$$

We first estimate the mean and then calculate the average squared deviation from the estimated mean.

Bias of the Maximum Likelihood Estimates

Imagine repeatedly drawing data sets of size N from the true distribution

$$\mathcal{N}(\mu, \sigma^2).$$

The maximum likelihood estimator of the mean satisfies

$$\boxed{\mathbb{E}[\mu_{\mathrm{ML}}] = \mu.}$$

Therefore, the sample mean is unbiased.
However,

$$\boxed{\mathbb{E}[\sigma^2_{\mathrm{ML}}] = \frac{N-1}{N}\sigma^2.}$$

Thus, the maximum likelihood variance systematically underestimates the true variance for finite N.

Why Is the Variance Underestimated?

The maximum likelihood variance is measured relative to

$$\mu_{\mathrm{ML}},$$

which was estimated from the same data.

The sample mean is chosen to minimize

$$\sum_{n=1}^{N}(x_n - \mu)^2.$$

Therefore, the observations are, on average, closer to the fitted mean than to the unknown population mean.

Intuition

Estimating the center from the data uses one degree of freedom, making the remaining spread appear slightly too small.



The Unbiased Sample Variance

The maximum likelihood variance is

$$\sigma^2_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}(x_n - \mu_{\mathrm{ML}})^2.$$

Multiplying it by $N/(N-1)$ removes the bias:

$$\boxed{\widetilde{\sigma}^2 = \frac{1}{N-1}\sum_{n=1}^{N}(x_n - \mu_{\mathrm{ML}})^2.}$$

Then

$$\boxed{\mathbb{E}[\widetilde{\sigma}^2] = \sigma^2.}$$

This explains why the usual sample variance uses $N-1$ rather than N.



Bias Becomes Smaller with More Data

The bias factor is

$$\frac{N-1}{N} = 1 - \frac{1}{N}.$$

As the sample size increases,

$$\frac{N-1}{N} \longrightarrow 1.$$

Therefore,

$$\sigma^2_{\mathrm{ML}} \longrightarrow \sigma^2 \qquad \text{as } N \to \infty.$$

For a simple Gaussian model, the bias is usually small unless N is small. In more complex models with many parameters, related estimation bias can lead to overfitting.

A Probabilistic View of Regression

In regression, we want to predict a target t from an input x.
Instead of assuming that the target is exactly equal to the model prediction, we allow Gaussian variation around it:

$$\boxed{p(t \mid x, \boldsymbol{w}, \sigma^2) = \mathcal{N}\left(t \mid y(x, \boldsymbol{w}), \sigma^2\right).}$$

Equivalently,

$$t = y(x, \boldsymbol{w}) + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

The function $y(x, \boldsymbol{w})$ specifies the conditional mean, while σ^2 describes uncertainty around that mean.

The Regression Likelihood

$$y(x_n, \boldsymbol{w}) = w_0 + w_1 x + w_2 x^2 + \cdots + w_M x^M = \sum_{j=0}^{M} w_j x^j.$$

Given training observations

$$\mathcal{D} = \{(x_i, t_i) : i = 1, \ldots, N\},$$

the likelihood is

$$\boxed{p(\boldsymbol{t} \mid \boldsymbol{x}, \boldsymbol{w}, \sigma^2) = \prod_{i=1}^{N} \mathcal{N}\left(t_i \mid y(x_i, \boldsymbol{w}), \sigma^2\right).}$$

The corresponding log likelihood is

$$\log p(\boldsymbol{t} \mid \boldsymbol{x}, \boldsymbol{w}, \sigma^2) = -\frac{1}{2\sigma^2} \sum_{i=1}^{N} \{t_i - y(x_i, \boldsymbol{w})\}^2 - \frac{N}{2} \log \sigma^2 - \frac{N}{2} \log(2\pi).$$



Gaussian Noise Leads to Squared Error

When maximizing the likelihood with respect to $\boldsymbol{w}$, the terms that do not depend on $\boldsymbol{w}$ can be ignored. Therefore,

$$\arg\max_{\boldsymbol{w}} \log p(\boldsymbol{t} \mid \boldsymbol{x}, \boldsymbol{w}, \sigma^2)$$

is equivalent to

$$\boxed{\arg\min_{\boldsymbol{w}} \frac{1}{2} \sum_{n=1}^{N} \{y(x_n, \boldsymbol{w}) - t_n\}^2.}$$

Interpretation

The sum-of-squares loss is not an arbitrary choice: it is the negative log likelihood implied by Gaussian observation noise.

Noise Variance and Predictive Distribution

After estimating $\boldsymbol{w}$, the maximum likelihood estimate of the noise variance is

$$\sigma^2_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N}\{y(x_n, \boldsymbol{w}_{\mathrm{ML}}) - t_n\}^2.$$

For a new input x, prediction is expressed as a distribution:

$$p(t \mid x, \boldsymbol{w}_{\mathrm{ML}}, \sigma^2_{\mathrm{ML}}) = \mathcal{N}\left(t \mid y(x, \boldsymbol{w}_{\mathrm{ML}}), \sigma^2_{\mathrm{ML}}\right).$$

The mean gives the point prediction, while the variance quantifies uncertainty around that prediction.



Quick Check

Suppose

$$\mathcal{D} = \{1, 2, 3\}.$$

Find the maximum likelihood estimates of μ and σ^2.

Quick Check

Suppose

$$\mathcal{D} = \{1, 2, 3\}.$$

Find the maximum likelihood estimates of μ and σ^2.

$$\mu_{\mathrm{ML}} = \frac{1+2+3}{3} = 2.$$

$$\begin{aligned}\sigma^2_{\mathrm{ML}} &= \frac{1}{3}\left\{(1-2)^2 + (2-2)^2 + (3-2)^2\right\} \\ &= \frac{2}{3}.\end{aligned}$$

The unbiased sample variance would instead be

$$\widetilde{\sigma}^2 = \frac{2}{3-1} = 1.$$



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Why Transform a Probability Density?

A change of variables allows us to construct a new random variable from an existing one.

$$X = g(Y) \qquad \text{or equivalently} \qquad Y = g^{-1}(X).$$

Examples include

$$Y = \log X, \qquad Y = X^3, \qquad Y = \frac{1}{1 + e^{-X}}.$$

Central question

If we know the density of one variable, how do we find the density of the transformed variable?

This calculation is the foundation of normalizing-flow models.



Ordinary Functions Transform by Composition

Suppose

$$x = g(y).$$

For an ordinary function $f(x)$, substitution gives a new function of y:

$$\boxed{\widetilde{f}(y) = f(g(y)).}$$

No additional factor is needed.

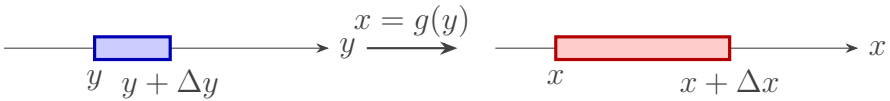
For a probability density, however,

$$p_y(y) \neq p_x(g(y))$$

in general, because a transformation can stretch or compress intervals.

Probability Mass Must Be Preserved

Suppose the small interval $(y, y + \Delta y)$ is mapped to $(x, x + \Delta x)$ by $x = g(y)$.



Both intervals represent the same event, expressed using different variables. Therefore, their probability masses must agree:

$$\boxed{p_y(y)\,\Delta y \approx p_x(x)\,\Delta x.}$$

The density must compensate for the change in interval width.

The One-Dimensional Change-of-Variables Formula

From conservation of probability mass,

$$p_y(y)\,|dy| = p_x(x)\,|dx|.$$

Therefore,

$$\boxed{p_y(y) = p_x(g(y))\left|\frac{dg(y)}{dy}\right|.}$$

The transformed density contains two components:

$$\underbrace{p_x(g(y))}_{\text{density at the mapped point}} \times \underbrace{|g'(y)|}_{\text{local change in scale}}.$$

Why Is There an Absolute Value?

The derivative $g'(y)$ can be negative when the transformation reverses direction.
For example, if

$$x = -2y,$$

then

$$\frac{dx}{dy} = -2.$$

But interval lengths and probability densities must be nonnegative. Therefore,

$$\boxed{\left|\frac{dx}{dy}\right| = 2.}$$

The absolute derivative measures the change in interval length, regardless of orientation.

Equivalent Formula Using the Inverse Transformation

Suppose the transformation is written as

$$y = f(x),$$

with inverse

$$x = f^{-1}(y).$$

Then

$$\boxed{p_y(y) = p_x\left(f^{-1}(y)\right)\left|\frac{df^{-1}(y)}{dy}\right|.}$$

Common mistake

When computing $p_y(y)$, use the derivative of the inverse mapping from y back to x, not $df(x)/dx$ directly.

Example: A Linear Rescaling

Suppose

$$X \sim \text{Uniform}(-2, 2)$$

and define

$$X = 2Y.$$

Because

$$p_x(x) = \frac{1}{4}, \qquad -2 < x < 2,$$

the density of Y is

$$\begin{aligned} p_y(y) &= p_x(2y) \left| \frac{d(2y)}{dy} \right| \\ &= \frac{1}{4} \times 2 = \boxed{\frac{1}{2}}, \qquad -1 < y < 1. \end{aligned}$$

The interval is compressed by a factor of two, so the density doubles.



Stretching and Compression

Probability mass must remain unchanged:

$$\text{density} \times \text{interval width} = \text{probability mass}.$$

Therefore:

- if a transformation stretches an interval, its density decreases;
- if a transformation compresses an interval, its density increases.

Stretching

Wider interval

$\Longrightarrow$

lower density

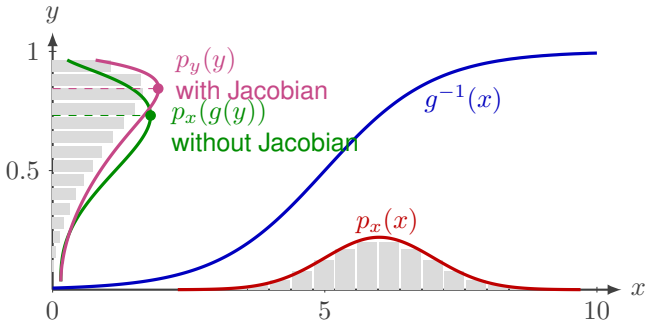
Compression

Narrower interval

$\Longrightarrow$

higher density

Nonlinear Transformation of a Density



The nonlinear transformation and its inverse are

$$x = g(y) = \log y - \log(1-y) + 5, \qquad y = g^{-1}(x) = \frac{1}{1+e^{-x+5}}.$$

The transformed density is

$$\boxed{p_y(y) = p_x(g(y)) \left| \frac{dg(y)}{dy} \right| = p_x\left(\log \frac{y}{1-y} + 5\right) \frac{1}{y(1-y)}.}$$

The Jacobian factor shifts the mode from the green curve to the magenta curve.

A Density Is Not an Ordinary Function

If an ordinary function transforms as

$$\widetilde{f}(y) = f(g(y)),$$

then a stationary point satisfies

$$\widetilde{f}'(y) = f'(g(y))g'(y) = 0.$$

Assuming $g'(y) \neq 0$, the location transforms in the expected way.
For a density,

$$p_y(y) = p_x(g(y))|g'(y)|,$$

so the derivative of the scale factor also affects the location of the mode.

Consequence

The numerical location of the maximum of a density depends on the variable in which the density is expressed.



Why Can a Nonlinear Transformation Move the Mode?

Assume

$$g'(y) = s|g'(y)|, \qquad s \in \{-1, 1\}.$$

Then

$$p_y(y) = s\, p_x(g(y))g'(y).$$

Differentiating gives

$$\boxed{p'_y(y) = s\, p'_x(g(y))\{g'(y)\}^2 + s\, p_x(g(y))g''(y).}$$

The second term,

$$p_x(g(y))g''(y),$$

arises from the local stretching or compression of the nonlinear mapping.
For a linear transformation, $g''(y) = 0$, so this additional term disappears and the mode transforms in the usual way.



Transforming Samples and Transforming Densities

There are two equivalent ways to obtain the distribution of $Y = f(X)$:

1. Transform many samples x_n into
$$y_n = f(x_n)$$
and examine their empirical distribution.
2. Transform the density analytically using the change-of-variables formula.

$$\boxed{p_y(y) = p_x(f^{-1}(y)) \left| \frac{df^{-1}(y)}{dy} \right|.}$$

If the derivative factor is omitted, the transformed curve will generally not match the distribution of the transformed samples.



Multivariate Change of Variables

Let

$$\boldsymbol{x} = g(\boldsymbol{y}), \qquad \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^D.$$

The multivariate change-of-variables formula is

$$\boxed{p_y(\boldsymbol{y}) = p_x(\boldsymbol{x}) \left|\det \boldsymbol{J}\right| = p_x(g(\boldsymbol{y})) \left|\det \boldsymbol{J}\right|,}$$

where $\boldsymbol{J}$ is the Jacobian matrix of g.

The absolute determinant replaces the absolute derivative from the one-dimensional formula.

The Jacobian Matrix

For

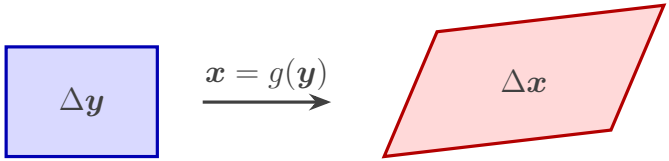
$$\boldsymbol{x} = g(\boldsymbol{y}),$$

the Jacobian matrix contains all first-order partial derivatives:

$$\boldsymbol{J} = \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{y}^\top} = \begin{pmatrix} \frac{\partial g_1}{\partial y_1} & \cdots & \frac{\partial g_1}{\partial y_D} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_D}{\partial y_1} & \cdots & \frac{\partial g_D}{\partial y_D} \end{pmatrix}.$$

Each entry describes how one output coordinate changes locally with one input coordinate.

The Determinant Measures Local Volume Change



Locally,

$$\boxed{\mathrm{Vol}(\Delta \boldsymbol{x}) \approx |\det \boldsymbol{J}| \, \mathrm{Vol}(\Delta \boldsymbol{y}).}$$

Because probability mass is preserved, the density changes by the same Jacobian factor.

Example: Two-Dimensional Rescaling

Consider

$$x_1 = 2y_1, \qquad x_2 = 3y_2.$$

The Jacobian is

$$\boldsymbol{J} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \qquad |\det \boldsymbol{J}| = 6.$$

Therefore,

$$\boxed{p_y(y_1, y_2) = 6\, p_x(2y_1, 3y_2).}$$

A unit area in $\boldsymbol{y}$-space becomes an area of six in $\boldsymbol{x}$-space, so the density in $\boldsymbol{y}$-coordinates is multiplied by six.



The Inverse Multivariate Formula

If the transformation is written as

$$\boldsymbol{y} = f(\boldsymbol{x}),$$

then

$$\boldsymbol{x} = f^{-1}(\boldsymbol{y}).$$

The transformed density is

$$\boxed{p_y(\boldsymbol{y}) = p_x(f^{-1}(\boldsymbol{y})) \left| \det \frac{\partial f^{-1}(\boldsymbol{y})}{\partial \boldsymbol{y}^\mathsf{T}} \right|}.$$

Equivalently,

$$p_y(\boldsymbol{y}) = p_x(\boldsymbol{x}) \left| \det \frac{\partial \boldsymbol{y}}{\partial \boldsymbol{x}^\mathsf{T}} \right|^{-1}.$$

Connection to Normalizing Flows

A normalizing flow begins with a simple density and applies a sequence of invertible transformations:

$$\boldsymbol{z}_0 \longrightarrow \boldsymbol{z}_1 \longrightarrow \cdots \longrightarrow \boldsymbol{z}_K = \boldsymbol{x}.$$

At each step, the log density changes by a log-determinant term. If

$$\boldsymbol{z}_k = f_k(\boldsymbol{z}_{k-1}),$$

then

$$\log p_k(\boldsymbol{z}_k) = \log p_{k-1}(\boldsymbol{z}_{k-1}) - \log \left| \det \frac{\partial f_k}{\partial \boldsymbol{z}_{k-1}^{\mathsf{T}}} \right|.$$

Summing across transformations gives

$$\boxed{\log p_x(\boldsymbol{x}) = \log p_0(\boldsymbol{z}_0) - \sum_{k=1}^{K} \log |\det \boldsymbol{J}_k|.}$$



Quick Check

Suppose

$$X \sim \text{Uniform}(0, 3)$$

and

$$Y = \frac{X}{3}.$$

Find the density of Y.

Quick Check

Suppose

$$X \sim \text{Uniform}(0, 3)$$

and

$$Y = \frac{X}{3}.$$

Find the density of Y.

$$x = 3y, \qquad \left|\frac{dx}{dy}\right| = 3,$$

$$p_y(y) = p_x(3y)\left|\frac{dx}{dy}\right| = \frac{1}{3} \times 3 = \boxed{1}, \qquad 0 < y < 1.$$

Thus,

$$Y \sim \text{Uniform}(0, 1).$$



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Information Theory

Probability describes uncertainty. Information theory quantifies how much information is revealed when uncertainty is reduced.

The main concepts are

- information content and entropy;
- differential and conditional entropy;
- Kullback–Leibler divergence;
- mutual information.

These concepts appear throughout machine learning in likelihood-based estimation, classification, representation learning, and generative modelling.



Information Content

Suppose a discrete random variable X takes the value x. The information received from observing x should satisfy:

- an unlikely event provides more information;
- a certain event provides no information;
- independent events provide additive information.

If X and Y are independent, then

$$p(x, y) = p(x)p(y)$$

and we require

$$h(x, y) = h(x) + h(y).$$

These properties lead to

$$\boxed{h(x) = -\log p(x).}$$



Information as Surprise

Using logarithms to base 2,

$$\boxed{h(x) = -\log_2 p(x).}$$

$p(x)$	$h(x)$	Interpretation
1	0 bits	certain event
$1/2$	1 bit	one binary decision
$1/8$	3 bits	three binary decisions
$1/1024$	10 bits	highly surprising event

Lower probability corresponds to greater information content.



Why Use a Logarithm?

For independent events,

$$p(x, y) = p(x)p(y).$$

The logarithm converts multiplication into addition:

$$\begin{aligned} h(x, y) &= -\log p(x, y) \\ &= -\log\{p(x)p(y)\} \\ &= -\log p(x) - \log p(y) \\ &= h(x) + h(y). \end{aligned}$$

The base determines the unit:

$$\begin{aligned} \log_2 &\Rightarrow \text{bits}, \\ \ln &\Rightarrow \text{nats}. \end{aligned}$$

The two units differ by the constant factor $\ln 2$.

Entropy

The entropy of a discrete random variable is its expected information content:

$$H[X] = \mathbb{E}[h(X)] = \sum_x p(x)\{-\log_2 p(x)\}.$$

Therefore,

$$\boxed{H[X] = -\sum_x p(x) \log_2 p(x).}$$

We use the convention $0 \log_2 0 = 0$, which follows from

$$\lim_{u \to 0^+} u \log_2 u = 0.$$

Entropy measures the average uncertainty before observing X.



Uniform Distribution and Coding Length

Suppose X has eight equally likely states:

$$p(x) = \frac{1}{8}.$$

Then

$$H[X] = -\sum_{x=1}^{8} \frac{1}{8} \log_2 \frac{1}{8} = -8 \left(\frac{1}{8}\right)(-3) = 3 \text{ bits}.$$

Three bits can represent the eight states:

$$000, 001, 010, 011, 100, 101, 110, 111.$$

The noiseless coding theorem states that entropy is a lower bound on the average number of bits required to encode the state.

Nonuniform Probabilities Allow Shorter Codes

Consider eight states with probabilities

$$\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}\right).$$

More probable states receive shorter codewords:

State	a	b	c	d	e	f	g	h
Code	0	10	110	1110	111100	111101	111110	111111
Length	1	2	3	4	6	6	6	6

The expected code length is

$$\frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{16}(4) + 4\left(\frac{1}{64}\right)(6) = 2 \text{ bits}.$$

This equals the entropy. A nonuniform distribution can be compressed more efficiently than a uniform distribution over the same states.

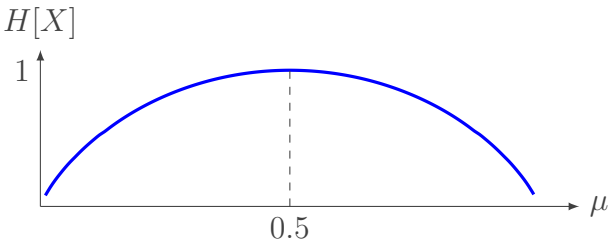
Entropy of a Bernoulli Variable

For $X \sim \text{Bernoulli}(\mu)$,

$$P(X = 1) = \mu, \qquad P(X = 0) = 1 - \mu.$$

Its entropy in bits is

$$\boxed{H[X] = -\mu \log_2 \mu - (1 - \mu) \log_2 (1 - \mu).}$$



The entropy is largest when both outcomes are equally likely.



Multiplicity and Entropy

Suppose N identical objects are distributed among bins, with n_i objects in bin i:

$$\sum_i n_i = N.$$

The number of compatible allocations is

$$\boxed{W = \frac{N!}{\prod_i n_i!}.}$$

W is the **multiplicity**. Entropy is proportional to its logarithm:

$$\boxed{H = \frac{1}{N} \ln W.}$$

A macrostate with many compatible microstates has high entropy.



Minimum and Maximum Discrete Entropy

For a discrete distribution over M states,

$$H[p] = -\sum_{i=1}^{M} p_i \ln p_i.$$

The minimum occurs when one state is certain:

$$p_j = 1, \qquad p_{i \neq j} = 0, \qquad H[p] = 0.$$

The maximum occurs for the uniform distribution:

$$p_i = \frac{1}{M}.$$

Therefore,

$$\boxed{0 \leq H[p] \leq \ln M.}$$

A sharply concentrated distribution has low entropy. A distribution spread evenly across its states has high entropy.

From Discrete to Continuous Entropy

Divide the real line into bins of width Δ. The probability of bin i is approximately

$$P_i = p(x_i)\Delta.$$

The entropy of the resulting discrete variable is

$$\begin{aligned} H_\Delta &= -\sum_i p(x_i)\Delta \ln\{p(x_i)\Delta\} \\ &= -\sum_i p(x_i)\Delta \ln p(x_i) - \ln \Delta. \end{aligned}$$

As $\Delta \to 0$, the first term approaches an integral. After removing the resolution-dependent term $-\ln \Delta$, we obtain

$$\boxed{H[X] = -\int p(x) \ln p(x)\, dx.}$$

This quantity is called **differential entropy**.

Differential Entropy Requires Care

Differential entropy resembles discrete entropy,

$$H[X] = -\int p(x) \ln p(x)\, dx,$$

but it has different properties.

- It depends on the measurement scale and coordinate system.
- It can be negative because a density can exceed one.
- It does not directly equal the number of bits needed to specify an exact real number.

For a uniform distribution on $(0, a)$,

$$p(x) = \frac{1}{a}, \qquad H[X] = \ln a.$$

Thus $H[X] < 0$ when $0 < a < 1$.

The Maximum-Entropy Problem

Suppose a continuous distribution has known mean and variance:

$$\int p(x)\,dx = 1, \qquad \int xp(x)\,dx = \mu,$$

$$\int (x-\mu)^2 p(x)\,dx = \sigma^2.$$

Among all densities satisfying these constraints, which has the largest differential entropy?
We maximize

$$-\int p(x)\ln p(x)\,dx$$

subject to the three constraints using Lagrange multipliers.
The stationary density has the form

$$p(x) = \exp\left\{-1 + \lambda_1 + \lambda_2 x + \lambda_3 (x-\mu)^2\right\}.$$

The Gaussian Has Maximum Entropy

Solving for the Lagrange multipliers gives

$$\boxed{p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}.}$$

Therefore, among all continuous distributions with a specified mean and variance, the Gaussian has the largest differential entropy.
Its entropy is

$$\boxed{H[X] = \frac{1}{2}\left\{1 + \ln(2\pi\sigma^2)\right\} = \frac{1}{2}\ln(2\pi e\sigma^2).}$$

A larger variance gives a larger entropy.



Cross-Entropy and Kullback–Leibler Divergence

Let $p(x)$ be the true distribution and let $q(x)$ be an approximation.
The cross-entropy of p relative to q is

$$H(p, q) = -\int p(x) \ln q(x)\, dx.$$

The additional coding cost from using q instead of p is

$$\begin{aligned} \mathrm{KL}(p\|q) &= H(p, q) - H(p) \\ &= \int p(x) \ln \frac{p(x)}{q(x)}\, dx. \end{aligned}$$

Thus,

$$\boxed{\mathrm{KL}(p\|q) = \int p(x) \ln \frac{p(x)}{q(x)}\, dx.}$$

KL Divergence Is Directional

In general,

$$\boxed{\mathrm{KL}(p\|q) \neq \mathrm{KL}(q\|p).}$$

For example, let

$$p = (0.9, 0.1), \qquad q = (0.5, 0.5).$$

Then

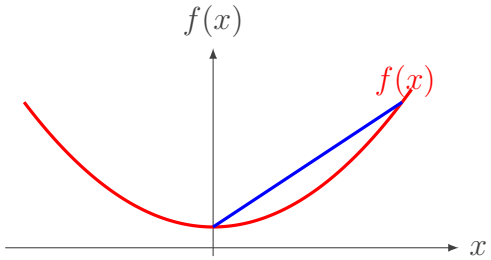
$$\mathrm{KL}(p\|q) = 0.9 \ln \frac{0.9}{0.5} + 0.1 \ln \frac{0.1}{0.5} \approx 0.368,$$
$$\mathrm{KL}(q\|p) = 0.5 \ln \frac{0.5}{0.9} + 0.5 \ln \frac{0.5}{0.1} \approx 0.511.$$

KL divergence is therefore not a distance metric.

Convexity and Jensen's Inequality

A function f is convex if every chord lies on or above its graph:

$$f\{\lambda a + (1-\lambda)b\} \leq \lambda f(a) + (1-\lambda)f(b), \qquad 0 \leq \lambda \leq 1.$$



For a convex function,

$$\boxed{f\big(\mathbb{E}[X]\big) \leq \mathbb{E}[f(X)].}$$

This is Jensen's inequality.



KL Divergence Is Nonnegative

Because $-\ln u$ is convex, Jensen's inequality gives

$$\begin{aligned}
\mathrm{KL}(p\|q) &= -\int p(x) \ln \frac{q(x)}{p(x)}\, dx \\
&\geq -\ln \left\{ \int p(x) \frac{q(x)}{p(x)}\, dx \right\} \\
&= -\ln \left\{ \int q(x)\, dx \right\} \\
&= 0.
\end{aligned}$$

Therefore,

$$\boxed{\mathrm{KL}(p\|q) \geq 0.}$$

Equality holds if and only if $p(x) = q(x)$ almost everywhere.



KL Divergence and Maximum Likelihood

Suppose $x_1, \ldots, x_N$ are drawn from an unknown distribution $p(x)$, and we approximate it using $q(x \mid \boldsymbol{\theta})$.

$$\begin{aligned}\mathrm{KL}\{p \| q_{\boldsymbol{\theta}}\} &= \mathbb{E}_p[\ln p(X)] - \mathbb{E}_p[\ln q(X \mid \boldsymbol{\theta})] \\ &\approx \frac{1}{N}\sum_{n=1}^{N} \ln p(x_n) - \frac{1}{N}\sum_{n=1}^{N} \ln q(x_n \mid \boldsymbol{\theta}).\end{aligned}$$

The first term does not depend on $\boldsymbol{\theta}$. Therefore,

$$\boxed{\arg\min_{\boldsymbol{\theta}} \mathrm{KL}\{p \| q_{\boldsymbol{\theta}}\} = \arg\max_{\boldsymbol{\theta}} \sum_{n=1}^{N} \ln q(x_n \mid \boldsymbol{\theta}).}$$

Maximum likelihood minimizes the forward KL divergence from the data distribution to the model.

Conditional Entropy

If X is known, the additional information needed to specify Y is

$$-\ln p(y \mid x).$$

Averaging over the joint distribution gives

$$\boxed{H[Y \mid X] = -\iint p(x,y) \ln p(y \mid x)\, dx\, dy.}$$

Using

$$p(x,y) = p(y \mid x)p(x),$$

we obtain the entropy chain rule:

$$\boxed{H[X,Y] = H[X] + H[Y \mid X].}$$

The joint uncertainty equals the uncertainty in X plus the remaining uncertainty in Y after X is known.



Conditional Entropy and Independence

If X and Y are independent, then

$$p(y \mid x) = p(y).$$

Consequently,

$$H[Y \mid X] = H[Y]$$

and

$$H[X, Y] = H[X] + H[Y].$$

If observing X helps predict Y, then the remaining uncertainty decreases:

$$H[Y \mid X] \leq H[Y]$$

for discrete random variables.
The reduction is measured by mutual information.

Mutual Information

If X and Y are independent, then

$$p(x, y) = p(x)p(y).$$

Mutual information measures the discrepancy between the joint distribution and the distribution implied by independence:

$$\begin{aligned} I[X, Y] &= \mathrm{KL}\big(p(x, y) \| p(x)p(y)\big) \\ &= \iint p(x, y) \ln \frac{p(x, y)}{p(x)p(y)} \, dx \, dy. \end{aligned}$$

Therefore,

$$\boxed{I[X, Y] \geq 0.}$$

Moreover,

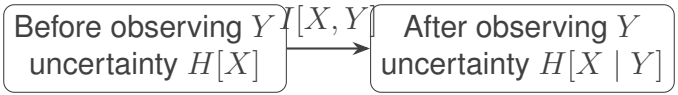
$$I[X, Y] = 0 \quad \Longleftrightarrow \quad X \text{ and } Y \text{ are independent.}$$

Mutual Information as Uncertainty Reduction

Mutual information has several equivalent forms:

$$\boxed{I[X,Y] = H[X] - H[X \mid Y] = H[Y] - H[Y \mid X].}$$

It measures how much observing one variable reduces uncertainty about the other.



From a Bayesian perspective, mutual information measures the expected reduction from prior uncertainty to posterior uncertainty after observing data.

A Simple Mutual-Information Example

Let X be a fair binary variable.

Perfect observation

If $Y = X$, then $H[X] = 1$ bit and $H[X \mid Y] = 0$. Therefore,

$$I[X, Y] = 1 \text{ bit.}$$

Independent observation

If Y is independent of X, then $H[X \mid Y] = H[X] = 1$ bit. Therefore,

$$I[X, Y] = 0.$$

Mutual information depends on statistical dependence, including nonlinear dependence. It is not restricted to linear association.